

Nanodroplet-Based Super-Resolution Ultrasound Localization Microscopy

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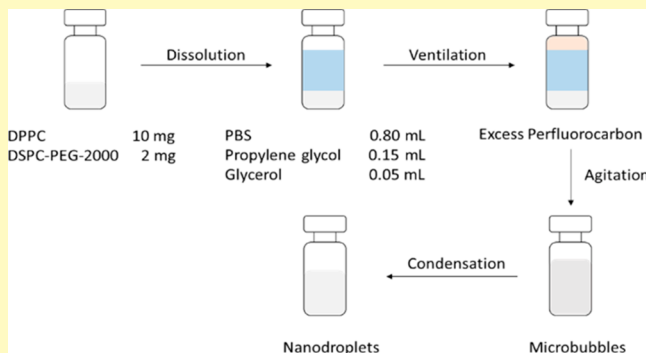
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ABSTRACT: Over the past decade, super-resolution ultrasound localization microscopy (SR-ULM) has revolutionized ultrasound imaging with its capability to resolve the microvascular structures below the ultrasound diffraction limit. The introduction of this imaging technique enables the visualization, quantification, and characterization of tissue microvasculature. The early implementations of SR-ULM utilize microbubbles (MBs) that require a long image acquisition time due to the requirement of capturing sparsely isolated microbubble signals. The next-generation SR-ULM employs nanodroplets that have the potential to significantly reduce the image acquisition time without sacrificing the resolution. This review discusses various nanodroplet-based ultrasound localization microscopy techniques and their corresponding imaging mechanisms. A summary is given on the preclinical applications of SR-ULM with nanodroplets, and the challenges in the clinical translation of nanodroplet-based SR-ULM are presented while discussing the future perspectives. In conclusion, ultrasound localization microscopy is a promising microvasculature imaging technology that can provide new diagnostic and prognostic information for a wide range of pathologies, such as cancer, heart conditions, and autoimmune diseases, and enable personalized treatment monitoring at a microlevel.

KEYWORDS: nanodroplets, microbubbles, ultrasound imaging, ultrasound localization microscopy, super-resolution, activation, recondensation, diagnosis, therapy



HIGHLIGHTS

- Ultrasound localization microscopy is a microvascular imaging technique that enables early diagnosis and monitoring of various diseases.
- Nanodroplets have the potential to speed up image acquisition time while ensuring image resolution.
- Ultrasound localization microscopy is required to offer a clear added value compared to the current clinical imaging techniques.

Optical localization microscopy (OLM), such as photo-activated localization microscopy (PALM) and stochastic reconstruction microscopy (STORM), has been demonstrated to have the capability to differentiate the plasma membranes and intracellular organelles at spatial resolutions below tens of nanometers.¹ The imaging mechanism of PALM is that the first laser pulse activates a subgroup of fluorescent proteins within the molecules. Then the second laser pulse is used to activate a new subgroup of fluorescent proteins.² After accumulating a stack of images, a spatiotemporal filter is applied to obtain the blinking fluorescent signals. Each spatially isolated fluorescent signal within every single image is localized as a localization event.³ All localization events from all of the

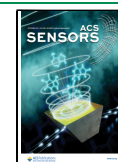
images are then accumulated into the final super-resolution image. OLM has revolutionized optical fluorescent imaging by imaging intracellular organelles below the diffraction limit.⁴ However, one of the major challenges of applying OLM is that organisms scatter the light significantly, which limits the penetration depth in biological tissues and makes in vivo imaging challenging.⁵

Compared with optical pulses, an acoustic wave is significantly less scattered in biological tissues in vivo and can be used to image deep structures. Microvascular changes in vivo are critical features in the visualization and characterization of various diseases such as tumor angiogenesis, peripheral arterial disease, coronary heart disease, etc. Gas-filled microbubble contrast agents are utilized in clinically routine contrast-enhanced ultrasound (CEUS) examinations to

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enhance blood flow signals.⁶ These microbubble contrast agents can generate strong resonant signals within the frequency range from 1 to 15 MHz compared with the tissue background, thereby enhancing the contrast of vasculature.⁷ The blood pool contrast agents remain in the blood vessels and do not spread into the extravascular space. They are not absorbed by any tissue or cell and are rapidly cleared from the blood through the pulmonary system.^{8,9} As a kind of pure blood pool contrast agent commonly used in clinics, it can provide basic information on relevant organs and vessels and visualize the degree of neovascularization of the lesions, which is helpful for the diagnosis and treatment of various diseases.¹⁰ However, the corresponding axial resolution is limited by the half-wavelength, and there is a compromise between imaging resolution and penetration depth. The penetration depth is determined by the decay of the ultrasound, which decreases with frequency while the resolution increases with frequency, so the deeper penetration depth indicates a lower resolution.¹¹ Inspired by OLM, super-resolution ultrasound localization microscopy (SR-ULM) was performed using a relatively low concentration of flowing microbubble contrast agents to visualize microvasculature beyond the image resolution limit.¹² The corresponding two-dimensional (2-D) and three-dimensional (3-D) SR-ULM techniques have shown promise in previous studies using microbubble contrast agents.

Because microbubbles are subwavelength, ULM requires the point spread function of a single microbubble for precise localization. Localization error occurs because of highly overlapping regions of the concentrated microbubbles, and a low concentration ensures that a single microbubble signal can be distinguished and further the localization of microbubbles can be achieved.¹³ Therefore, microbubble-based SR-ULM requires relatively low concentrations of flowing microbubbles. Flow is crucial here because sparse activation of microbubbles is not possible, unlike the case for OLM. Such a requirement for microbubble relocation or replenishment significantly increases ULM acquisition time, especially for microvasculature with relatively slow flow rates, such as capillary networks, while nanodroplets are independent of flow rate.^{14,15}

To overcome this challenge, previous studies have shown that a relatively high concentration of low-boiling-point nanodroplets can be randomly and sparsely activated within the clinical safety limits and then deactivated as required by controlling the applied ultrasound amplitude, thus demonstrating the ultrasonic counterpart of PALM at depth.^{16,17} Nanodroplets are converted from liquid to microbubbles under acoustic or laser activation, and the resulting spatially stationary, temporally transient microbubbles will realize the formation of SR-ULM before recondensation into a liquid nanodroplet state.¹⁸ Nanodroplets are disrupted by interleaved activation and imaging pulses. This enables the rapid sampling of different subpopulations of microbubbles between successive imaging frames. This may reduce the acquisition time and is unaffected by flow and concentration.¹⁶ In addition to this, repeated activation and recondensation of high-boiling-point nanodroplets are possible, which demonstrates the feasibility of nanodroplet-based SR-ULM.¹⁹

This article reviews various nanodroplet-based ultrasound localization microscopy techniques and their corresponding imaging mechanisms. In addition, we summarize the preclinical application and the challenges required to be overcome to promote the clinical translation of nanodroplet-based ultra-

sound localization microscopy. Finally, the future perspective of this technique is discussed.

■ ULTRASOUND CONTRAST AGENTS

CEUS imaging is sensitive to tissue vasculature as the ultrasound contrast agents flow through. These contrast agents could highlight the microvasculature, as the ultrasound transducer is also sensitive to slowly moving microbubbles. With the rapid developments in the design of these ultrasound contrast agents, nanodroplet contrast agents can be manufactured via condensation and extrusion methods.²⁰ Compared with optical contrast agents, ultrasound contrast agents can travel deep within the tissues, which makes them advantageous in clinical contrast imaging modalities.

Microbubbles. Microbubble contrast agents have now developed to the third generation.²¹ According to the different gas cores within the microbubbles, they are mainly air, nitrogen, carbon dioxide, fluorine sulfur gas, fluorine carbon gas, etc.²² The first-generation ultrasound microbubble contrast agent has no shell and is mainly represented by air-filled microbubbles.²³ Their stabilities are relatively weak, as they could not travel far within the vasculature after intravenous injection. Additionally, the *in vivo* circulation time is relatively short, which results in limited clinical applications. For example, the diffusion of the contrast agent Levovist in blood is faster than that of the previous product. The former is air-based, where small molecules can easily diffuse into the bloodstream through microbubbles.²⁴ A number of attempts have also been made by previous studies, such as manufacturing shells with polymer-based materials. However, these contrast agents failed to be commercialized due to poor imaging quality.²⁵ Perfluorinated gas with low solubility in water replaced air as the second-generation microbubble contrast agent, and the shell material began to transform into liposomes, but it did not have specific targeting.²⁶ The third-generation ultrasound contrast microbubbles are drug-loaded microbubbles with targeted function, which are used for diagnosis and treatment of diseases.^{27,28} Microbubbles with gaseous cores encapsulated by lipid shell layers are the most commonly used contrast agents in clinical CEUS examinations.²⁹ Microbubble contrast agents normally have a diameter ranging from 1 to 10 μm .³⁰ Therefore, they cannot travel outside the vasculature as blood pool contrast agents.

At present, the commercialized microbubble contrast agents approved by the FDA mainly include Sonovue (Bracco, Milano, Italy), Sonazoid (GE Healthcare, WI, USA), and Definity (Lantheus Medical Imaging, MA, USA).²⁶ Chen et al. performed preoperative lymphography enhanced ultrasound in patients with thyroid carcinoma using the microbubble contrast agent SonoVue to evaluate the ultrasound features of lymph nodes, the area of lymphatic drainage, and the detection of sentinel lymph nodes.⁸ Lahtinen et al. used the Sonazoid contrast agent to perform CEUS of superficial lymphatic vessels in the upper extremities of humans.³¹ It is potentially valuable in assessing the kinetics of lymph fluid and allowing imaging of abnormal lymphatic anatomy. Karalko et al. applied carbon dioxide foam to ultrasound imaging of a rat heart to achieve high ultrasound echoes. It minimizes tissue damage compared to iodine contrast agents.³² At the same time, microbubbles are the most typical precursors of nanodroplets, which need to be further processed into nanodroplets through mechanical stirring, pressurization,

Table 1. Activation Conditions and Situations of Various Nanodroplets in Key Studies

core	concentration (droplets/mL)	temp (°C)	size (nm)	mechanical index	number of cycles	ref
octafluoropropane	1.0×10^9	37	350	0.2	1	58
decafluorobutane	1.8×10^8	37	119	1.3	1–2	16
perfluorohexane, perfluorobutane	2.0×10^9	37	202–420	1.1–1.7	2–4	19
octafluoropropane	3.3×10^9	24	153	0.8	1	45

condensation, and other steps.³³ Researchers have shown that commercial and FDA-approved Definity (Lantheus Medical Imaging, Billerica, MA, USA) could be used to produce nanodroplets. The nanodroplets used by Lea-Banks are formed by Definity through agitation, pressurization, and other steps. It was proved to be a tool for neurosuppression and nerve stimulation.³³ Sheeran et al. directly applied the clinical contrast agent Definity to form nanodroplets through mechanical agitation.³⁴ With the rapid development of nanomedicine, nanobubbles can be novel contrast agents. Nanobubbles, usually formed from lipid shells and perfluorocarbon gas cores, are 10 times smaller than the microbubbles. Nanobubbles are compared with microbubbles and usually have a low contrast signal but are widely used in the treatment of various malignancies.³⁵ Nanobubbles have the potential to be extravasated into the tumor space due to their enhanced permeability and retention (EPR) effect. These extravasation potentials contribute to highly sensitive observations of tumor biomarkers, which are important for early cancer diagnosis.^{36,37}

Nanodroplets. Recently, nanodroplet (ND) contrast agents have been widely investigated, as these nanodroplets could offer more flexibility in ultrasound imaging compared to microbubbles. Nanodroplets normally have a perfluorocarbon liquid core and are encapsulated by a lipid shell.³⁸ The advantages of using nanodroplet contrast agents in ultrasound imaging are summarized below. First, it has been found in a previous study that nanodroplet contrast agents have significantly longer half-lives in vivo compared to microbubble contrast agents. The microbubble half-life is usually 3–5 min, and the average half-life of nanodroplets is 30–60 min.³⁹ Second, nanodroplet contrast agents can be spatiotemporally activated into microbubble contrast agents to selectively provide ultrasound contrast signals, which offers more flexibility in real-time ultrasound imaging.⁴⁰ Due to the requirements for sparsity of flowing microbubbles, the microbubble-based ULM requires a long acquisition time, especially for small microvasculatures.⁴¹ Since not all nanodroplets are activated at the same time, activating nanodroplets detected by ultrasound can be achieved at very high agent concentrations, with or without flow.⁴² When it comes to cancer imaging, nanodroplet-based ULM has the potential to offer enhanced insights into tumor angiogenesis, characterized by the presence of vessels with very slow flows.⁴³ Nanodroplets are invisible to ultrasound before activation, and when nanodroplets are acoustically or optically activated, they form transient microbubbles that immediately exhibit hyperechogenicity in ultrasound imaging, and as droplets are activated and deactivated at the ultrasound pulse repetition frequency, the signal can accumulate as quickly as sending imaging pulses, enabling fast cumulative localization, resulting in faster super-resolution imaging.^{44–46} Third, the nanodroplets can extravasate into the cancerous space due to the leaky vasculature of cancerous endothelial walls and the EPR effect, which make them useful in cancer extravasation imaging.^{35,47,48} Many studies have tried to verify that nanodroplets can extravasate

within a range of sizes, usually with an optimum exosmosis size of 100–300 nm.^{49,50} Rapoport et al. performed experiments on mouse thigh subcutaneous muscle and adipose tissue under a microscope, observed the vasculature, and found that the extravasation rate of nanodroplets into the normal tissue was very slow.⁵¹ Song et al. have demonstrated that cavitation-facilitated permeability was enhanced across the blood–brain barrier in rats induced by acoustically vaporized nanodroplets.⁵² This indicates that phase change nanodroplets can be used as a safe, efficient, and durable reagent to achieve satisfactory cavitation-mediated permeability enhancement in biomedical applications. Due to the limited capacity to load therapeutic agents and short in vivo circulation time, it is difficult for microbubble contrast agents to achieve effective drug concentration in a tumor site. However, previous studies have demonstrated that drug-loaded nanodroplets could be aggregated in the tumor site by enhancing the permeability and retention effects. Additionally, it could open the blood–brain barrier to achieve drug delivery in the brain. These nanodroplets can be converted into microbubbles under targeted ultrasound irradiation and release therapeutic drugs within the microbubbles.^{33,38} This alleviated some of the cytotoxic effects of conventional chemotherapeutic agents on healthy cells. Therefore, drug-loaded nanodroplets could provide better tumor treatment and imaging methods due to the advantages of strong drug-carrying capacity, long half-life, and the ability to reach out the extravascular space.⁵³ Ho et al. applied doxorubicin-loaded nanodroplets and ultrasound to treat tumor-bearing mice, which improved the therapeutic efficacy.⁵⁴ Lea-Banks et al. injected decafluorobutane nanodroplets loaded with drug-mimicking dye into rats and exposed their brains to ultrasound. The dye was found to penetrate through the blood–brain barrier and was retained locally in the brain tissue. Therefore, this proved the capability of drug-loaded nanodroplets to open the blood–brain barrier and deliver drugs.⁵⁵

Finally, compared to microbubbles, nanodroplets can selectively enhance and observe the blood supply of a single vessel of interest and its downstream vessels.⁴⁰ Because nanodroplets need to be activated and deactivated by ultrasound, nanodroplets with different boiling points require different acoustic energies. In many previous studies, nanodroplets were classified according to their boiling points. Xu et al. investigated and compared the cavitation characteristics of flowing low- and high-boiling-point phase-shift nanodroplets during focused ultrasound exposures.⁵⁶ Meanwhile, McHugh et al. pointed out that the cavitation of nanodroplets with a high boiling point requires high acoustic pressures while the nanodroplets with a low boiling point will be converted into microbubbles at lower acoustic pressures.⁵⁷ So this review will classify nanodroplets according to boiling point. We present the activation conditions and situations of various types of nanodroplets in Table 1.

Low-Boiling-Point Nanodroplets. Low-boiling-point nanodroplets normally include octafluoropropane (C₃F₈,

Table 2. Characteristic Parameters of Various Nanodroplets Used in Different Nanodroplet-Based Ultrasound Localization Microscopy

chemical formula	name	molecular wt (g/mol)	boiling point (°C)	spatial resolution (μm)	temporal resolution (Hz)	activation conditions	application	ref
C ₆ F ₁₄	perfluorohexane (PFH)	338	56	7–16	several hundred	laser	imaging of the mouse brain	61
C ₆ F ₁₄ , C ₄ F ₁₀	perfluorohexane (PFH), decafluorobutane (DFB)	338, 238	56, −2	64	100	acoustic	imaging of the rabbit quadriceps muscle	19
C ₄ F ₁₀	decafluorobutane (DFB)	238	−2	40	1000	acoustic	visualization of rabbit renal microvasculature	17
C ₃ F ₈ , C ₄ F ₁₀	octafluoropropane (OFP), decafluorobutane (DFB)	188, 238	−37, −2	115	500–10000	acoustic	imaging of the rabbit renal microvasculature	42

OFP)-based and decafluorobutane (C₄F₁₀, DFB)-based nanodroplets as described in previous studies.⁵⁹ As can be seen from Table 2, OFP and DFB have relatively lower boiling points compared to perfluorohexane (C₆F₁₄, PFH).⁶⁰ Therefore, they have a relatively lower activation threshold and thus they can be activated into microbubble contrast agents via diagnostic ultrasound pulses at depth.⁴⁵ It was found in the previous literature that the activation threshold for activating an OFP nanodroplet is 0.14 mechanical index (MI) whereas that for activating a DFB nanodroplet is 0.40 MI.²⁰

A low-boiling-point nanodroplet can be manufactured by condensing and pressurizing microbubble contrast agents. Briefly, a vial containing the microbubbles was kept submerged in an ice-salt bath and then pressurized with ambient air into the vial, as shown in Figure 1. The advantage of using low-

were used to assisted in the activation of the high-boiling-point nanodroplets.^{61,65–67} Luke et al. used laser energy to activate high-boiling-point perfluorocarbon nanodroplets to image the brains of a mice that have undergone craniotomy.⁶¹ Yoon et al. used laser energy to activate perfluorohexane nanodroplets for super-resolution imaging.⁶⁷ Zhang et al. used acoustic energy to activate octafluoropropane nanodroplets for ultrasound super-resolution images.⁶⁸ However, the advantage of using high-boiling-point nanodroplets compared to the low-boiling-point nanodroplets is that they are significantly more stable at the physiological temperature.⁶⁹

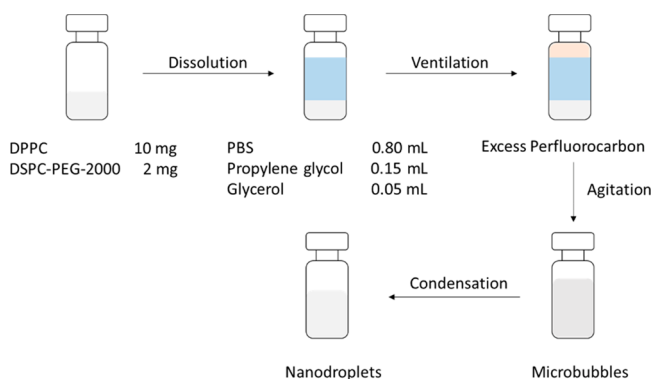
■ PRINCIPLES OF ULTRASOUND LOCALIZATION MICROSCOPY

Principle and Trilemma of Microbubble-Based ULM.

For microbubble-based SR-ULM, low concentrations of microbubbles are required for the localization processing. Then a dynamic video of microbubble flow is obtained in either the B-mode or contrast mode. At that point a crucial part of the ultrasound localization processing is to distinguish microbubbles from the surrounding tissue. A clutter filtering technique was used to enhance the contrast signals and suppress the background tissue signals. Once these filtered contrast signals were obtained, these spatially isolated signals were localized. As the point spread function of the imaging system can define the acoustic response to a single isolated point scatterer, the coordinates of each signal can be identified, and its accuracy is significantly higher than that of diffraction-limited resolution.¹⁴ After localization processing, these localized signals between the consecutive frames can be tracked to define the paths and velocity of microbubbles within the microvasculature.

In practice, microbubble-based SR-ULM imaging requires a compromise of imaging speed, fidelity, and resolution, as shown in Figure 2. With the improvement of image acquisition speed, the image resolution of microvasculature will be sacrificed. The improvement of microvascular image resolution will affect the degree of image expression of microvasculature.

Injection of low-concentration microbubbles can achieve better resolution but a longer imaging time.⁷⁰ The image acquisition can be done faster by injecting a higher concentration of MBs, which sacrifices the resolution.⁷¹ Another option is just acquiring for a shorter period of time using a low MB concentration while achieving a good resolution; however, this reduces the fidelity of the final SR images and the diagnostic quality.^{72,73} In conclusion, MB-based SR imaging has the trilemma given in Figure 2.^{41,74} Any method, algorithm, or technique that breaks or bends this triangle is a big step forward in SR-ULM.^{75,76}

**Figure 1.** Manufacturing process of low-boiling-point nanodroplets.

boiling-point nanodroplets compared to high-boiling-point nanodroplets is that the low-boiling-point nanodroplet can be relatively easily activated to microbubble contrast agents via acoustic pulses within a clinical safety level.^{16,45,62–64} However, as a lower-boiling-point liquid core is used, spontaneous vaporization may happen, as it is not very stable at the physiological temperature. It was found in a previous study that the spontaneous vaporization of low-boiling-point nanodroplets may help with ULM processing, as it could potentially provide additional localizations.⁴⁵

High-Boiling-Point Nanodroplets. A previous study has demonstrated that the application of high-boiling-point nanodroplets (perfluorohexane, C₆F₁₄) in SR-ULM.⁶¹ Compared to the low-boiling-point nanodroplets (decafluorobutane, C₄F₁₀), perfluorohexane-based nanodroplets have a significantly higher activation threshold; therefore, they require much higher energy to be activated to microbubbles. In a number of previous studies, both acoustic and photo energies

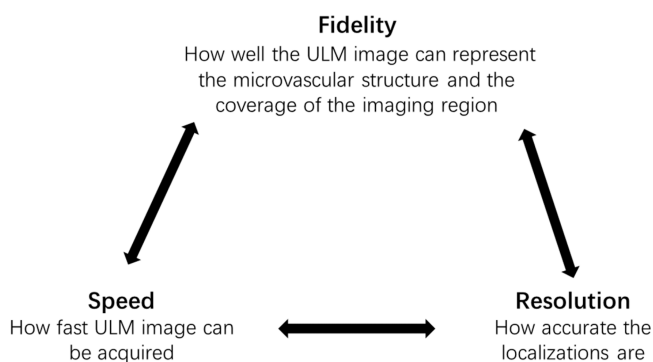


Figure 2. Compromise among fidelity, speed, and resolution in the generation of SR-ULM.

Nanodroplets are the arguably the best solution to this trilemma, as they can be injected in high concentration and activated stochastically on demand, which therefore accelerates the image acquisition speed while covering the whole imaging region.⁴² Since the activations of nanodroplets are performed in a stochastic fashion and only a small subset of nanodroplets are activated at a given time, a high imaging resolution can be achieved through accurate localizations.

Principle and Application of Low-Boiling-Point Nanodroplet-Based ULM. There are several schemes for low-boiling-point nanodroplet-based ULM such as acoustic wave sparsely activated localization microscopy (AWSALM), fast-AWSALM, localization of the acoustic droplet vaporization signal, etc. We present two of them here, AWSALM (Figure 3)

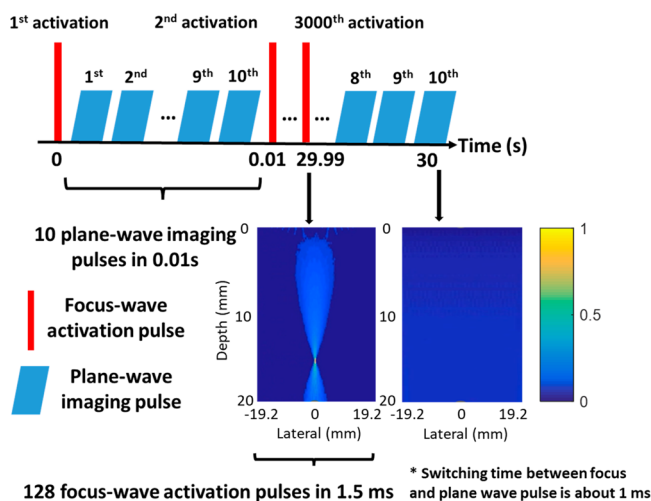


Figure 3. Illustration of AWSALM ultrasound pulse sequence.

and fast-AWSALM (Figure 4). The basic imaging mechanism of AWSALM is that a subgroup of nanodroplets is sparsely activated by the first focus-wave activation pulse, because only the most easily activated nanodroplets are activated by selective activation pulse pressure.¹² Until the next activation pulse, these activated nanodroplets will produce acoustic signals. First, the existing activated nanodroplets generated by the previous activation pulse will be destroyed;⁶⁸ Second, a new subgroup of nanodroplets will be activated.⁴² The localizations of different subgroups of activated nanodroplets can be offered by this continuous activation and subsequent imaging pulses. Zhang et al. prepared decafluorobutane nanodroplets and verified that AWSALM utilizes acoustic

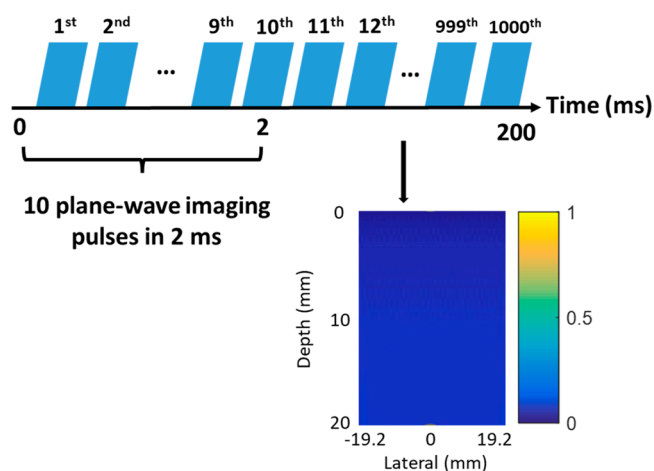


Figure 4. Illustration of Fast-AWSALM ultrasound pulse sequence.

waves to sparsely and stochastically activate nanodroplets by acoustic vaporization and to simultaneously deactivate the existing vaporized nanodroplets via acoustic destruction in a crossed-tube phantom.¹⁶ Their team also extended the application of AWSALM to an animal; the in vivo ULM of rabbit kidney microvessels was obtained within 1.1 s.¹⁷ Zhang et al. developed fast-AWSALM with low-boiling-point octafluoropropane nanodroplets and high-frame-rate plane waves for simultaneous activation, destruction, and imaging to achieve ULM on a subsecond time scale.^{45,66} As for AWSALM technology, the technology does not require flow.

The imaging mechanism of fast-AWSALM is different from that of AWSALM, as fast-AWSALM utilizes plane-wave pulses for both activation and imaging of nanodroplets.⁶⁶ Due to the lower activation threshold of low-boiling-point nanodroplets, the acoustic pressure provided by the plane-wave imaging pulse is sufficient to activate it for imaging.^{45,64} With plane waves, faster imaging and activation acquisition can be achieved without using focus-wave. Since merely the most easily activated droplets are activated by selected acoustic pressure, the subgroup of octafluoropropane nanodroplets will be sparsely activated by the first plane-wave imaging pulse, which is the principle of fast AWSALM.⁴² These activated droplets will generate contrast signals until the next plane-wave pulse. There are three purposes for using these image pulses. The first purpose is to generate images, the second purpose is that a large number of the existing microbubbles activated by the previous pulse will be destroyed, and the third purpose is that a new subgroup of droplets will be activated into microbubbles. This continuous imaging, activation, and destruction pulses enable different localizations of various subgroups of activated droplets within the region of interest.

Principle and Application of High-Boiling-Point Nanodroplet-Based ULM. A previous study has demonstrated the use of high-boiling-point nanodroplets for ULM. Nanodroplets are sparsely and randomly activated by optical or acoustic means to reconstruct fine structures of the tissues. These high-boiling-point nanodroplets are made of a perfluorohexane liquid core encapsulated by a lipid shell and photoabsorbers to enable optical triggering.⁷⁷ Due to the incorporation of the fluorosurfactant shell and near-infrared dye, these nanodroplets can be optically activated repeatedly.⁶⁵ Nanodroplets with a perfluorocarbon-filled core and a shell of lipid and Pluronic F68 can undergo reversible vaporization and

reliquefaction when driven by clinically safe acoustic pulses (MI: 1.1–1.7). Therefore, a reversible phase transition can occur between the liquid and gaseous states to achieve multiple activations. Dong et al. has shown that using these high-boiling-point nanodroplets can reconstruct a ULM image of the nonflow tube phantom.¹⁹ Therefore, the spatially isolated and temporally transient microbubbles offer high acoustic signals before they recondense to their original liquid state. The recondensation of high-boiling-point nanodroplets is a stochastic process. This means that at any given time only a sparse subset of nanodroplets goes through the gas-to-liquid transition. This process may be propelled by a combination of the particle size, laser energy, amount of encapsulated dye, local pressure, and ambient temperature. Therefore, as long as the imaging frame rate is sufficiently high, the response of an individual nanodroplet can be isolated by obtaining the difference between adjacent frames.⁶⁷ All the responses can be localized and then summed up to generate the final ULM image.¹⁶

Dong et al. used three criteria for the implementation of SR-ULM using a kind of blinking acoustic agent. First, the processes of activation and recondensation of these nanodroplets require reversibility, and the process of activation requires it to be purely controlled by acoustic pulses from the same ultrasound imaging transducer used for imaging acquisition. Second, the acoustic pressure for the nanodroplet activation requires it to be below the diagnostic limit. Third, the sparse and stochastic activation and recondensation signals require achieving a high reconstruction efficiency for SR-ULM.¹⁹ These nanodroplets overcome the main limitations of localization imaging based on microbubbles so that vascular imaging, especially small vessels, is no longer confused by slow blood flow. In addition, it only needs 100 frames of ultrasound imaging, which is 1 order of magnitude less than that of the conventional SR-ULM and is enough to reconstruct the ULM without flow through cumulative localization or temporal radial autocorrelation. Compared to the low-boiling-point nanodroplets, the purpose of this study is to detect the spatially isolated recondensation events for localizations. Additionally, this technique does not rely on flow. However, the activation of a number of high-boiling-point nanodroplets (e.g., perfluorohexane nanodroplets) requires the assistance of laser pulses, as their activation thresholds are relatively high compared to those of other nanodroplets with lower boiling points.⁶¹ This limits the penetration depth since the laser pulse can only penetrate a few millimeters in tissue.

■ PRECLINICAL AND CLINICAL APPLICATIONS OF ULM

Early Detection of Cancer. Early detection of cancer found by medical imaging methods, while tumors are still small and before metastasis, will greatly increase the chances of successful treatment.^{78,79} Key information about the efficacy of the therapeutic approach is offered by imaging during treatment. Due to the low cost, wide accessibility, and safety, ultrasound has the latent capacity to turn into the preferred method of cancer imaging.⁸⁰ Doppler ultrasound techniques or conventional B-mode are challenged to demonstrate microvascular characteristics on the scale of early tumor angiogenesis. ULM is not limited to diffraction-limited resolution, which makes this technique a potentially forceful technique.^{81,82} Lin and his research team successfully imaged the tumor microvasculature of rat subcutaneous fibrosarcoma

tumors using ULM and evaluated tumor-related angiogenesis. Compared with conventional ultrasound, the resolution was improved by 10 times.⁸³ This technology can be used to detect tumor-related angiogenesis, even though in deep tissues, malignancies can be distinguished by their angiogenic fingerprint.⁸⁴ In subsequent studies, Lin et al. observed 3-D microvascular patterns of rat subcutaneous fibrosarcoma tumors with ULM and at the same time compared the microvascular characteristics with the control healthy tissues.⁷⁸ The curvature of blood vessels in tumor-bearing mice is significantly higher than that in the control group, and the degree of vascular heterogeneity is higher. This technology has the potential to distinguish between healthy and diseased tissues and detect tumor-related angiogenesis. The changes in lymph node microcirculation often indicate metastasis. The identification and quantification of metastatic lymph nodes are crucial for tumor prognosis and treatment. Zhu et al. performed ULM on rabbit popliteal lymph nodes and realized the quantification of microvascular structure and blood flow dynamics of lymph nodes.⁸⁵ ULM can not only provide tumor microvascular information but also measure the hypoxia state of tumors. Lowerison et al. successfully imaged xenogeneic renal cell carcinoma growing on the chorioallantoic membrane of chicken embryo using ULM and evaluated the microvascular structure, vascular perfusion, and hypoxia state of the tumor. They found that the microvascular signal in the center of the tumor was generally lower than that around the tumor and showed a corresponding high degree of hypoxia probe expression, which was not obvious in conventional ultrasound contrast processing. It was found that the vessel curvature was significantly correlated with the quantity of hypoxia probe.⁸⁶ Zhang et al. successfully imaged human thyroid nodules and breast masses with ULM and obtained useful clinical information such as microvascular flow rate and microvessel density, which is helpful for the differential diagnosis of benign and malignant.^{87,88} Different types of tumors can be distinguished by the fine details of vascular network provided by ULM. Opacic et al. used motion model ULM to image breast cancer mice with different vascular phenotypes and extracted new super-resolution imaging parameters, including relative blood volume (rBV), blood flow direction and velocity, and distance between blood vessels. They evaluated and verified these parameters with high-resolution microcomputed tomography (micro-CT) scanning and histological analysis of tumor slices, proving that the technology can identify tumors with different vascular phenotypes.⁸⁹ In addition, ULM has great potential in monitoring the response of early tumors to drug therapy. Ghosh et al. imaged breast-cancer-bearing mice before and after administration and found that there was an acute microvascular response within 2 h after administration, which was verified with the results of immunohistochemistry. They also evaluated the longitudinal changes in the tumor microvascular network in the response to vascular targeted drug therapy.⁹⁰

Tissue Characterization. Nanodroplet-based ULM is currently used for study in vitro and in animals since there are no commercial nanodroplet contrast agents approved by the FDA.⁹¹ With the development of nanodroplet-based ULM, this imaging technique can be used to display the microvasculature of various organs in vivo with faster imaging acquisition.⁹² Riemer et al. demonstrated that ULM showed diastolic and systolic perfusion in the microvasculature of rabbit kidney by using sono-switchable nanodroplets and

Table 3. Summary of Key Research on Ultrasound Localization Microscopy in Disease Diagnosis and Monitoring

number	object	model	application	ref
1	subcutaneous fibrosarcoma	rat	evaluation of tumor-related microvasculature	78
2	popliteal lymph nodes	rabbit	quantification of microvasculature and blood flow dynamics	83
3	renal cell carcinoma xenograft model	chicken embryo	evaluation of hypoxia status of tumor	86
4	thyroid nodules	human	differentiation of benign and malignant	87
5	breast cancer	mice	differentiation of tumors with different vascular phenotypes	89
6	breast cancer	mice	monitor the early response to drug treatment	90
7	atherosclerotic plaque	rabbit	provision of neovascular density and related characteristics	94
8	deep ocular tissue	rabbit	evaluation of microvessel and blood flow velocity	95
9	skeletal muscle	type 2 diabetes mice	evaluation of microvasculature structure and parameters	96

found that the spatial resolution was improved by 4 times.⁴² Luke et al. used high-boiling-point perfluorocarbon nanodroplet-based as a “blinking” contrast agent to image the brain microvasculature of a craniotomy mouse and found that an axial resolution of 8 μm and a lateral resolution of 16 μm can be achieved in vivo, which is significantly better than that of conventional ultrasound imaging.⁶¹ Dong et al. used laser-activated high-boiling-point nanodroplet-based ULM to show the microvasculature of quadriceps femoris in the hind leg of Japanese white rabbits.¹⁹ The result demonstrated that laser-activated nanodroplet-based ULM in image reconstruction reduced the number of frames by 1 order of magnitude than that in conventional ULM. Zhang et al. displayed the microvascular structure of rabbit kidney with the AWSALM technology based on the activation and deactivation of nanodroplets.¹⁷ The result showed that the in vivo super-localization image can be obtained in 1.1 s using AWSALM.

Diagnosis of Vascular Diseases. Changes in microvascular structure, flow rate, and other indicators often indicate the emergence of diseases, such as major diseases related to diabetes, atherosclerosis, and so on. Changes in microvascular indicators often precede changes in large blood vessels. Early monitoring and evaluation of the microvascular system facilitates early diagnosis and treatment of disease.⁴³ ULM has broken through the acoustic diffraction limit, significantly improved the spatial resolution, and is able to evaluate the early changes of microvasculature, thus improving the ability of disease monitoring and diagnosis.⁹³ Chen et al. successfully identified the rabbit femoral artery and its surrounding neovascularization using ULM. They observed the density of neovascularization and its related characteristics in the attachment of atherosclerotic plaque in rabbits and performed micro-CT imaging and histopathological and morphological verification.⁹⁴ Qian et al. used the ULM technology to image the deep tissue of rabbit eyes with high resolution, evaluated the microvasculature and flow velocity, and detected the subtle changes of blood flow. This has great potential in detecting and monitoring eye diseases such as glaucoma in the future.⁹⁵ Ghosh et al. successfully used ULM to image the skeletal muscle microvasculature of type 2 diabetes mice and compared them with lean mice to evaluate the structure and parameters of microvasculature, which is a new method for measuring the characteristics of skeletal muscle microvasculature and the dysfunction of type 2 diabetes.⁹⁶ In addition, ULM has also been used to evaluate the changes in renal microvasculature. Chen et al. successfully imaged the renal microvasculature of mice 21 and 42 days postischemia-reperfusion injury using ULM and noninvasively quantified the changes of renal microvasculature in mice, which has great potential for future evaluation of progressive renal diseases.⁹⁷ Lowerison et al.

successfully imaged the brain microvasculature of aging mice by using the technology of ULM, providing the microvascular structure and function information on the cerebral microvasculature. They also quantified the differences in cerebral vascularity, blood velocity, and vessel tortuosity across several brain regions. Compared with young mice, the blood flow velocity and the blood volume of the cerebral cortex of aging mice was significantly reduced, while the blood vessel curvature was significantly increased.⁹⁸ These findings show that ULM technology has great potential in disease diagnosis and monitoring, as shown in Table 3. In addition to the aforementioned research in microvascular diseases in animals, the ULM technique was also applied to image the carotid artery, lower limbs, liver, kidney, pancreas, and other organs in humans to prove the feasibility of performing ULM in deep tissues in humans.^{99–102} Due to limitations such as long acquisition time and out-of-plane movement, further technical improvements are required to achieve better clinical results and a wider range of applications.

Detection and Diagnosis of Arthritis. Microvascular endothelial dysfunction is an early and/or important event in the development of cardiovascular disease and related organ damage and also exists in patients with rheumatoid arthritis (RA).¹⁰³ Fluorescent-dye-labeled hyaluronic acid is fixed on the surface of nanoparticles and can be used as a nanoprobe and contrast agent to selectively detect target molecules and diagnose the progress of RA.^{104–106} In addition, due to the infiltration of abnormal blood vessels and inflammatory cells, the vascular permeability of the RA site was significantly improved. Through the EPR effect, nanoparticles infiltrated into synovial tissue through the gap between endothelial cells and slowly released drugs.¹⁰⁷ The combination of passive targeting and EPR effect can maximize the effect of drug treatment.¹⁰⁸ The specific ligand recognized by cells at an RA site is combined with the receptor on the surface of nanoparticles to selectively deliver the drug to the expected action site while further reducing the drug retention in normal tissues.¹⁰⁹ Compared with traditional therapeutic drugs, it has less toxicity and adverse side effects.¹¹⁰ Although ULM has not been applied to the detection of microvasculature related to RA, it is believed that nanodroplet-based ULM has the potential to provide key information for the diagnosis and treatment of RA.

DISCUSSION

ULM can provide comprehensive and quantitative microvascular parameters noninvasively and has great value in monitoring the diagnosis and treatment of microvascular diseases, such as the diagnosis and efficacy evaluation of tumors, chronic kidney diseases, inflammation, arteriosclerosis,

rosis, diabetes, etc.¹¹¹ ULM using contrast agents has displayed its tremendous latent capacity.

Previous studies have discussed different types of ULM based on nanodroplets, which can be activated by a laser or ultrasound, respectively. For example, Luke et al. developed a transient laser-activated nanodroplet-based ULM and used it to image the brains of mice. The nanodroplets could be located within several micrometers, which is beneficial to achieve high-resolution molecular imaging at substantial depth in tissue.⁶¹ Dong et al. introduced a type of blinking acoustic nanodroplets, which could achieve ULM imaging at quite high concentrations and have much higher efficiency of repetitive activation.¹⁹ Tang and his team developed nanodroplet-based AWSALM and fast-AWSALM and further verified the ability to visualize microvasculature in vivo and in vitro, respectively, which opens up the possibilities for super-resolution molecular imaging.^{16,17,42} Nevertheless, there are a few challenges to be conquered to speed up the clinical translation of ULM. Out-of-plane motion, long acquisition time, and low frame rates are the major challenges faced by the clinical translation of ULM.^{43,112–114} A previous study has attempted to overcome out-of-plane motion by implementing a 3D imaging technique.⁷⁸ The microbubble-based ULM results in a long image acquisition time due to differentiation of individual microbubble signals. However, the nanodroplet-based ULM can significantly shorten image acquisition time by sparsely activating nanodroplets and deactivating nanodroplets as required.¹⁶ Despite the fact that the super-resolution imaging quality may be poor due to the low imaging frame rate on clinical imaging systems, there are increasingly more new-generation ultrasound systems that could support high-frame-rate imaging.¹⁰⁰

However, with all of these challenges, ULM has also been implemented on patients. Although these challenges may lead to imaging which are less accurate, the results in previous studies have shown the significance of this technique.^{87,88} Due to the lack of a gold standard, verifying the accuracy of super-resolution ultrasound imaging (SRUS) in reliable clinical applications remains a well-recognized challenge. At present, several studies have applied a number of techniques (confocal microscopy, two-photon microscopy, micro-CT, etc.) to evaluate the accuracy of ULM in the detection of microvascular structure. In vivo, the quality of the super-resolved images is often evaluated through the self-coherence of the vessel branching pattern.⁴³ Christensen-Jeffries et al. compared optical images with super-resolution images of the same plane, where shallow microvascular networks were observed. To determine the accuracy of ULM in the shallow tissues, it can be compared to established microscopy techniques such as confocal microscopy or two-photon microscopy.¹¹⁵ Opacic et al. compared the level of tumor vascularization obtained by mULM to rBV values obtained by maximum intensity over time (MIOT) postprocessing, ex vivo micro-CT, and immunohistochemical (IHC) analysis of tumor sections. The average speed obtained by motion model ULM was compared with the average speed calculated by the complementary dynamics. The quantitative values of the distance (mean, variance and maximum) to the nearest vessel obtained by IHC and mULM were compared by them.⁸⁹ Several studies have used micro-CT and histopathological examination to verify the authenticity of ULM microvascular detection.^{97,116}

A recent study in the field of SR-ULM is concerned with the development of machine learning algorithms. Due to the

overlap of high concentration microbubbles, noise, tissue artifacts, and motion in the image, SR-ULM image quality and spatial resolution can be reduced and localization error can be increased.^{13,117} Applying machine learning can identify and reject nonsingle microbubble echoes and filter artifacts and noise in the images. In nanodroplet-based SR-ULM techniques, nanodroplets will activate into microbubbles at a lower MI and bring them to a moderate concentration. Therefore, applying machine learning will bring the same filtering, detection, and localization effects to the activated nanodroplets.¹¹⁸

One of the challenges for nanodroplet-based ultrasound localization microscopy is that all the nanodroplet contrast agents used in the previous literature were manufactured in the lab. This means none of them have gone through clinical trials. This is due to the fact that the nanodroplet contrast agents have not been FDA approved because they may induce biological effects. A previous study has demonstrated that the activation process of smaller nanodroplets may cause damage to the surrounding tissues, because the acoustic pressure and temperature required to activate the droplets increase as the particle size decreases.³⁸ The clinical translation of nanodroplet-based ULM still requires effort. To this day, the application of ultrasound localization microscopy has been confirmed in a number of preclinical studies and different animal models. Nanodroplet contrast agents with different liquid cores have been developed so that the corresponding activation mechanism can be comprehended. Nevertheless, a better normalization of activation strategies is expected to accelerate the clinical translation of nanodroplet-based SR-ULM. This is because the development of clinical-grade ultrasound contrast agents demands large-scale and extensive clinical trials to prove the safety. Testing the biological effects of nanodroplet activation by hematology and histology found that no biological effects caused by nanodroplet activation were observed in the case of low MI.¹¹⁹ For nanodroplet contrast agents, they still require large-scale clinical trials. This is to prove its applicability in the human body, so as to make further efforts for the acceleration of clinical translation. In addition, the academic research of newly discovered nanodroplet contrast agents faces a challenge, which is to generally supply industrial support for Phase II and III clinical trials. This makes it harder to narrow the gap in clinical translation.

Another challenge is that the current activation strategies reported in the literature may be difficult to integrate into the clinical ultrasound system. A number of studies utilized laser energy to aid activating nanodroplets. However, it is challenging to integrate laser emission into the clinical ultrasound examination routine, and this may also make the imaging procedures more complicated for ultrasound sonographers. Most of the studies utilized the “Imaging + Activation” ultrasound pulse sequences to obtain activated droplet signals. However, the current clinical ultrasound systems do not have the corresponding pulse sequences to support the activation of nanodroplets, which requires further improvements. The last concern about the nanodroplet-based ULM is the safety issue regarding the acoustic energy required to activate the nanodroplets. The FDA-approved mechanical index (MI) value of 1.9 is the maximum threshold for diagnostic imaging. Currently, the MI used in all ultrasound super-resolution studies is within the range defined by the FDA, and the acoustic pressure range is 0–5 MPa.^{16,19,45,68} Although this technique has been widely investigated in preclinical studies, a

number of challenges still exist. For instance, out-of-plane motion, long acquisition time, low frame rates, and no clinically available nanodroplets are difficulties in the clinical translation of nanodroplet-based ULM at present. However, with continuous investigations and efforts, it is believed that this technique may potentially be utilized in clinical diagnosis.

CONCLUSION

Driven by the novel design of advanced imaging sequences, development of high-frame-rate ultrasound systems, and ultrasound contrast agents, ultrasound localization microscopy has demonstrated its tremendous latent capacity in many clinical and preclinical studies by expanding the application of conventional ultrasound imaging techniques. It has been shown that the application of these various ultrasound contrast agents in early cancer diagnosis and sonodynamic therapy is feasible. With the rapid development of molecular chemistry, imaging processing techniques, and ultrasonic physics, ultrasound localization microscopy is expected to become a highly sensitive and specific imaging tool. This will help clinicians better diagnose and manage diseases. Nevertheless, in order to be widely accepted by clinicians, these ultrasound localization microscopy techniques need to provide distinguishable added value compared with current clinical ultrasound imaging techniques. Therefore, academic researchers and pharmaceutical companies need to jointly promote the translational process to accelerate clinical translation.

In summary, nanodroplet-based SR-ULM was developed to produce ultrasound super-resolution images on a subsecond scale. This is several orders of magnitude faster than the existing microbubble-based SR-ULM. Compared with microbubble contrast agents, the size of nanodroplets is significantly smaller. This indicates that this technique can potentially resolve the structure, which is beyond the scope of current microbubble-based SR-ULM approaches.

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Notes

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VOCABULARY

ultrasound imaging, a medical imaging technique using high-frequency sound waves to produce echoes in the body; ultrasound contrast agent, a solution that can enhance blood flow signals after intravenous administration; ultrasound localization microscopy, an imaging technique for breaking the ultrasound diffraction limit to display the microvasculature within tissues by locating and tracking the ultrasound contrast agent; microbubble contrast agent, an ultrasound contrast agent filled with a gas core; nanodroplet contrast agent, a phase change ultrasound contrast agent, usually with a perfluorocarbon liquid core and encapsulated in a lipid shell; activation, the process of converting nanodroplets from liquid droplets to gaseous bubbles under acoustic or optical emissions.

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